2(a)(i)	**M1** nuclear charge increases (across period) **AND** similar shielding **M2** greater / stronger attraction to nucleus	**2**
2(d)(i)	(thermal stability) increases (down the group)	**1**
2(d)(ii)	NO_2 **AND** O_2	**1**

Examiner report missing
9701_w25_qp_23 Question 2

2(a)(i)	delocalised electrons move (through the structure / between layers)	**1**
2(e)(iii)	**M1** (mixture / solution becomes more) yellow **M2** (forward) reaction is exothermic (equilibrium shifts left) **OR** equilibrium / reaction shifts (left) to absorb (additional) heat / energy **M3** (mixture / solution becomes more) red **M4** fewer moles of gas on right-hand side **OR** (gas on left and) no gas on the right	**4**
2(f)(i)	$MgCl_2(aq) + Na_2CO_3(aq) \rightarrow MgCO_3(s) + 2NaCl(aq)$ **M1** formulae in balanced equation **M2** state symbols	**2**
2(f)(ii)	MgO / magnesium oxide **AND** CO_2 / carbon dioxide	**1**
2(f)(iii)	(thermal stability) increases (down the group)	**1**

Examiner report missing
9701_w25_qp_22 Question 2

2(a)(i)	**M1** nuclear charge increases (across period) **AND** similar shielding **M2** greater / stronger attraction to nucleus	**2**
2(d)(i)	(thermal stability) increases (down the group)	**1**
2(d)(ii)	NO_2 **AND** O_2	**1**

Examiner report missing
9701_w25_qp_21 Question 2

2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	**2**
2(b)(i)	ionic radius increases / ions larger (down the group) extra shell (of electrons)	**2**
2(d)(i)	*barium oxide* (white) solid disappears / dissolves *barium carbonate* effervescence **AND** (white) solid disappears / dissolves	**2**
2(d)(ii)	$Sr + 2H_2O \rightarrow Sr(OH)_2 + H_2$	**1**
2(d)(iii)	decreases (down the group)	**1**

Examiner report missing
9701_m25_qp_22 Question 2

3(a)(i)		**3**

element	C (graphite)	Sn
state and appearance	grey shiny solid	silvery solid
electrical conductivity	● good/ conductor	good
type of bonding	● covalent	metallic
type of structure	giant	● giant

3(b)(i)	$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$	**1**

3(b)(ii)	thermal stability **increases** down the group	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as MgCl and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiCl_4 is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid SiCl_4 as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiCl_4 molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and SiCl_4 and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

3(a)(i)	(natural =)lightning	**1**
	(man-made =)internal combustion engines	**1**

3(a)(iv)	$2HNO_3 + CaO \rightarrow Ca(NO_3)_2 + H_2O$	**1**

Question 3

(a) (i) Many candidate answers referred to lightning as responsible for one natural occurrence of NO_2. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

(ii) Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of NO_2 as a catalyst in the production of sulfur-based acid rain.

(iii) Candidates who were familiar with this area of the syllabus described how PAN forms from NO_2. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with NO_2.

(iv) Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of H_2 rather than H_2O.

(v) Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

(b) (i) Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

(ii) The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

(iii) Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

(iv) Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

(c) (i) Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

(ii) A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

(d) Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

(e) Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

(f) (i) This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in P_2 identified as sp^2 or sp^3
- in P_2 overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

(ii) Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

3(a)(i)		**3**

element	C (graphite)	Sn
state and appearance	grey shiny solid	silvery solid
electrical conductivity	● good/ conductor	good
type of bonding	● covalent	metallic
type of structure	giant	● giant

3(b)(i)	$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid $SiCl_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid $SiCl_4$ molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and $SiCl_4$ and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

1(a)(i)	**M1** (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond **M2** (N_2 molecules /molecule(s) of N_2) are non-polar	**2**

1(d)(i)	(+)IV	**1**

1(d)(ii)	thermal decomposition	**1**

Question 1

This question assessed inorganic chemistry, with some aspects of physical chemistry. This tested principally in the AO1 category. Candidates showed some facility of recall but were equally divided in response when asked to apply their knowledge to unfamiliar circumstances.

(a)(i) This standard recall question could be answered with easy reference to syllabus points.

(ii) With reference to the syllabus, orbital overlap can be directly quoted; knowledge of the bonding in N_2 is also straightforward, so the number of σ and π bonds should be readily accessible.

(b)(i) Many candidates answered this question well.

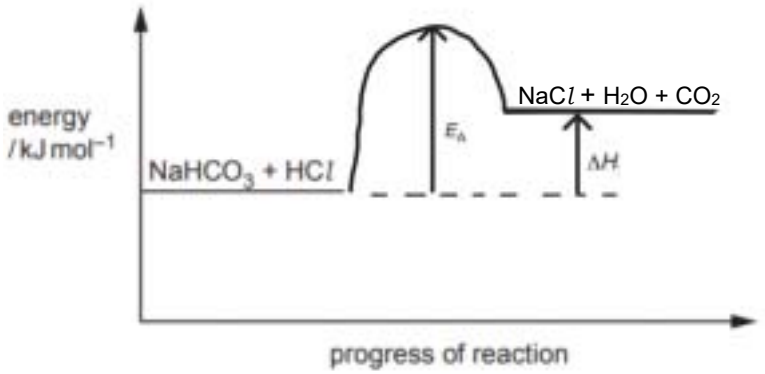
(ii) This question gave rise to many variations of answer. Candidates are advised to be brief with their answers: often, correct responses were contradicted by (incorrect) explanations of valence electrons.

(c)(i) Many candidates answered this question well.

(ii) The reaction of P_4O_{10} was better known than that of Al_2O_3. The latter is not less common but requires better honed balancing skills if constructed rather than written in recall.

(d)(i) Many candidates answered this question well.

(ii) Many candidates were able to identify the type of reaction that occurs.

2(a)	 *All three correct (2 marks) or two correct (1 mark):* • profile for endothermic reaction (include 'hump') • arrow representing E_A • arrow representing ΔH **OR** (+) 27.2	**2**
2(c)	$Ca(NO_3)_2 \rightarrow CaO + 2NO_2 + ½O_2$ **M1** correct formula for calcium nitrate: $Ca(NO_3)_2$ **M2** correct balanced equation	**2**

Question 2

(a) Appropriate profiles for an endothermic reaction were common. In general, representation of activation energy and enthalpy change were correct. Some answers showed very imprecise location of arrows on the diagram or showed arrows which did not point in the correct direction.

(b) (i) Good answers described an appropriate observation based on the idea that no more gas is produced when the reaction stops.

(ii) This calculation proved to be demanding. The specific heat capacity of water was not always given to the appropriate number of significant figures. The step to convert the amount of energy produced in the experiment to the amount of energy produced per mol was often confused. A significant proportion of answers included a negative sign for the enthalpy change value to reflect the exothermic nature of the reaction. Some answers were given in terms of J mol^{-1} rather than kJ mol^{-1}, as directed in the question.

(iii) The best answers showed a correct Hess' cycle in their working. Weaker responses tended to use appropriate stoichiometry in their calculation but confused the algebraic relationship between ΔH_1 and ΔH_r in the calculation. A relatively high proportion of candidates did not attempt to answer this question.

(c) Many answers deduced that **Z** was a Group 2 nitrate. Identification of calcium as a Period 4 element presented some difficulties and $Sr(NO_3)_2$ was seen on occasions. Production of the balanced equation for this thermal decomposition proved to be demanding.
The weakest responses identified the brown gas as bromine and gave equations of Group 2 bromides producing bromine.

2(a)(i)	increases down the group	**1**
2(a)(ii)	white solid disappears	**1**
2(a)(iii)	CO_2 / carbon dioxide	**1**
2(a)(v)	$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$	**1**
2(b)(i)	$CaCO_3$ **AND** thermal stability of carbonates **increases** down the group	**1**
2(b)(ii)	$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

(iii) In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, CO_2, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

(iv) The addition of ethanoic acid to $Ba(OH)_2$ gave barium ethanoate, $(CH_3COO)_2Ba$, as the product.

Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

(v) $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

(b) (i) The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

(ii) The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

2(a)	curved line from (0,0) to reach same horizontal line	**1**
2(f)(i)	magnesium hydroxide / $Mg(OH)_2$	**1**
2(f)(ii)	$Mg(NO_3)_2 \rightarrow MgO + 2NO_2 + \frac{1}{2}O_2$	**1**
2(f)(iii)	increases	**1**

Question 2

(a) Some candidates appeared to confuse graphs of rate versus time with those showing concentration versus time.

(b) Well-rehearsed arguments related the significance of increasing temperature on the position of equilibrium to the example given. More able candidates showed no ambiguity and made explicit reference to the effect of an increase in temperature on the position of this equilibrium in terms of either the '*forward reaction is exothermic'* or '*the reverse reaction is endothermic'*. Less precise answers included '*the reaction moves backwards as it's an exothermic reaction'*.

(c) This question proved to be demanding. A few candidates found K_p using an appropriate method; they worked out the mole fraction of the components, converted them into partial pressures and then substituted these values into the correct expression for K_p. Weaker candidates calculated K_p directly from moles and included moles in the units. Deduction of units also presented difficulties. Many answers gave units in terms of Pa with no attempt to convert kPa to Pa within the calculation.

(d) Lightning was well known as a natural process responsible for formation of atmospheric NO and NO_2. 'Catalytic converter' as the man-made process was a common incorrect answer.

(e) (i) Many correct definitions of the term free radical were seen. Ambiguous descriptions referred to '*unpaired electron'* as '*unshared*' or '*unbonded*' or '*free*'. Common incorrect answers referred to a free radical as *'a species with a lone pair of electrons'*.

(ii) The pair of equations which showed the catalytic role of NO_2 in the oxidation of SO_2 were not well known. The equation to represent the regeneration of NO_2 appeared to cause most difficulties with many incorrectly balanced equations seen.

(iii) The environmental consequence of the oxidation of sulfur dioxide was well known.

(f) (i) Identification of a magnesium containing base was relatively common. Incorrect answers included salts, rather than bases, which contained magnesium or bases that were not made from magnesium. The formula of the compound identified was not always correct.

(ii) This equation proved challenging for many. Some answers incorrectly balanced equations which included all correct species. Use of an incorrect formula for magnesium nitrate was a common error, including Mg_3N_2.

(iii) Identification of the change in thermal stability down Group 2 nitrates was fairly well known.

2(a)(i)	increases down the group	**1**
2(a)(ii)	white solid disappears	**1**
2(a)(iii)	CO_2 / carbon dioxide	**1**
2(a)(v)	$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$	**1**
2(b)(i)	$CaCO_3$ **AND** thermal stability of carbonates **increases** down the group	**1**
2(b)(ii)	$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$	**1**

Question 2

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$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

2(a)(i)	it / aluminium oxide is insoluble (in water) **OR** it / aluminium oxide does not react (with water)	**1**

2(b)	$Mg(NO_3)_2$	**1**

Question 2

(a) (i) Many candidates knew that aluminium objects covered in a layer of aluminium oxide do not react with cold water because the aluminium oxide is insoluble in water and does not react with the water.

(ii) The equation, with state symbols, was quite well answered, with many candidates recognising that the product was aqueous sodium aluminate.

(iii) Some good answers were seen.

(b) Some candidates gave the formulae of nitrates of elements that are not located in Period 3.

(c) (i) Several candidates correctly stated the oxidation number of Al in $AlCl_3$. The answers for that of P in PCl_5 were varied.

(ii) Many candidates knew that Al and P had different numbers of electrons available to use. Many omitted where these electrons are located.

(iii) It was well recognised that PCl_5 was the required chloride of phosphorus and that HCl is a product of this reaction. Few candidates gave the correct product containing phosphorus. Many answers contained an oxide of phosphorus and/or an incorrect phosphoric acid.

(iv) Some candidates knew the reaction is a hydrolysis reaction with other candidates correctly describing it as a substitution reaction.

(v) Some good answers were seen.

2(a)	**M1** (enthalpy change) when 1 mole of water is formed **M2** from an (aqueous) acid and an alkali / (aqueous) base	**2**

2(c)(i)	..(1)..$H_2SO_4(aq)$ + ..(1)..$Ba(OH)_2(aq)$ → …(1)..$BaSO_4(s)$ + $2H_2O(l)$…… **M1** correct state symbols for both products **M2** correctly balanced equation	**2**

Question 2

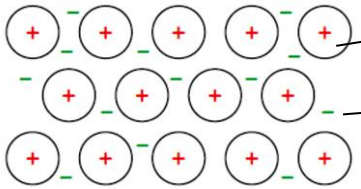
(a) Some candidates were able to give a concise and accurate definition. Overall, the recall of this definition was not well attempted. Candidates often omitted any reference to a molar scale, although 'one mole of acid reacting with one mole of base' was commonly seen as an inadequate response. A common misconception was that neutralisation requires energy.

(b) (i) Few candidates were able to incorporate the relevant stoichiometry involved into their calculations.

(ii) Very few candidates gained full credit here. A very small proportion of candidates correctly used 175 g as the correct mass, many instead quoting 1 g. Many candidates converted the temperature to Kelvin but erroneously adding 273 to 7.8. A very small proportion of candidates correctly divided the energy released by the number of moles of water formed.

(c) (i) This was well answered by most candidates. Most were able to provide a correct balanced equation. Some gave the incorrect state of barium sulfate.

(ii) Many candidates stated that $BaSO_4$ is insoluble, though not many were able to link this to an additional energy change. Several answers incorrectly referred to the solubility of $Ba(OH)_2$.

2(a)(i)	 **M1** diagram showing minimum of 4 particles (in total in 2 rows) • circles containing X^{n+} do not have to be labelled •  must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} **AND** with the circles surrounded by electrons shown as e^- / – **OR** little circles labelled electrons	**1**
	M2 label / legend showing **delocalised** electrons	**1**
2(b)	$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$	**1**
2(c)	reagent = any named/formula of soluble sulfate OR H_2SO_4 OR any named/formula of soluble hydroxide	**1**
	$BaSO_4$ insoluble (& $MgSO_4$ soluble) OR ($Ba(OH)_2$ soluble &) **$Mg(OH)_2$ insoluble**	**1**

Question 2

This was an inorganic-chemistry question based around Group 2.

(a) (i) A labelled diagram showing the structure of a metal crystal was required here. Candidates are reminded of the subtle difference between delocalised and free electrons.

(ii) There were many good answers here although some responses continued to use free electrons instead of delocalised.

(b) A large proportion of candidates offered the reaction with steam ($\rightarrow MgO + H_2$) rather than that with cold water, although there were many good answers.

(c) Several answers gave examples of metals which would 'displace' Mg but not Ba. Water was a favourite reagent, with the explanation given as either difference in solubility or a hydrolysis type reaction. Such answers were not sufficiently creditworthy.

(d) (i) This question was answered quite well by candidates.

(ii) A good proportion of candidates understood what was needed, but many spoilt a good answer by stating that X = Ba. The most common error was not to include the stoichiometric factor of ½: such answers tended to have further inconsistency in the subtraction of the A_r of $1\times$ or $2 \times Cl$.